DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 08 OF 12

Sustainability Concept & Strategy

Water, carbon, energy and shade — stated conservatively, including where the scheme runs a deficit.

CONTENTS

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	**github.com/wasim437/AI_SAFA**
Live portal	**wasim437.github.io/AI_SAFA/**
Produced by	**src/plan.py** · **src/costing.py** · **src/climate.py**
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Al Safa 2 Park — Falaj Al Safa

Mohamed Wasim · Individual Applicant

Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

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THE CODE AND DATA BEHIND THIS DOCUMENT

src/plan.py
src/costing.py
src/climate.py

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py          # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline   # 38 checks
```

Sustainability Concept & Strategy

Water, carbon, energy and shade — including where the scheme runs a deficit

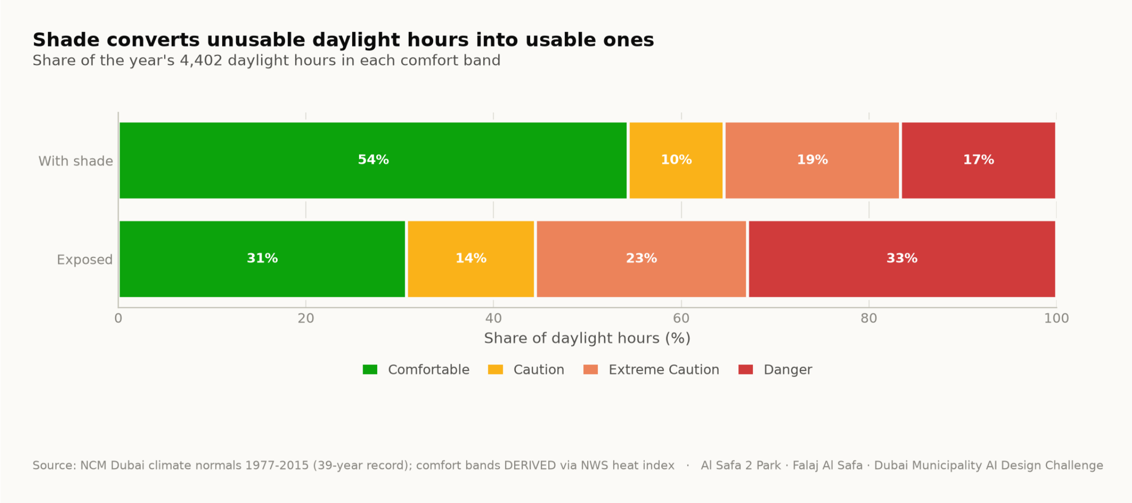
A sustainability chapter that reports only its successes is not evidence, it is marketing. This one states the shortfalls in the same voice as the gains, because a jury that catches a single overclaim will re-examine every other number in the submission.

Upload slot 08 — Sustainability Concept & Strategy
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Thermal performance — the primary outcome

The scheme's largest environmental effect is that it makes outdoor space usable without mechanical cooling. Comfortable daylight hours rise from 44.5% to 64.6%; the mean heat-index reduction under the canopy is 7.13 °C; and peak heat index falls from 56.8 °C to 48.7 °C.



Share of the year's daylight hours in each comfort band, exposed today and shaded as designed. Analysis output — computed from project data.

2. Water

Al Falaj is a 0.9 m recirculating channel — about 105 m² of water surface in total. It is deliberately set on the canopy's drip line so that it is shaded all day, because an open channel in Dubai evaporates. **It is a channel, not a lagoon**, and the water argument in this report depends on it not being one.

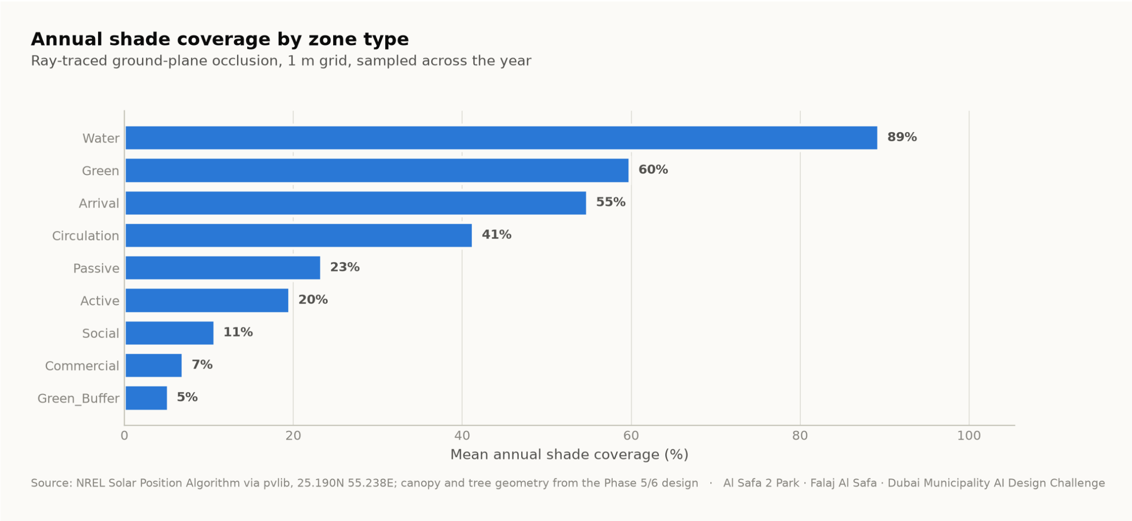
- Irrigation is subsurface drip fed from treated sewage effluent, not potable supply.
- Planting is five desert species selected for drought tolerance; Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.
- The falaj recirculates and is shaded for its whole length.

3. Energy — stated as the deficit it is

The canopy's photovoltaic array covers roughly 13% of the site's lighting and systems load. That is a deficit, not a surplus. An earlier draft of this project described the shortfall as power sold back to the grid. It is corrected here. The park is a net consumer of electricity. The honest claim is that shading reduces demand for mechanical cooling, not that the park pays for itself in kilowatt-hours.

4. Carbon, biodiversity and materials

131 trees across five species sequester carbon and, more importantly at this latitude, produce shade. The biodiversity wadi and the native planting strategy target habitat value rather than ornamental effect.



Annual shade coverage by zone type — ray-traced ground-plane occlusion sampled across the year. Analysis output — computed from project data.

5. What is conservative, and what is assumed

- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively — the real shaded condition is slightly better than reported, never worse.
- **The 6 °C shade relief is a literature value**, not a site measurement.
- **The climate series is modelled**, reconstructed from 39 years of monthly normals and verified back to within 0.39 °C. Conclusions are about a typical year, never about extremes.
- **Site-wide mean shade is 34.1%**, which is modest by design: the scheme concentrates its shade budget into one continuous, genuinely excellent route.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Fig02 comfort bands

Analysis output — computed from project data.

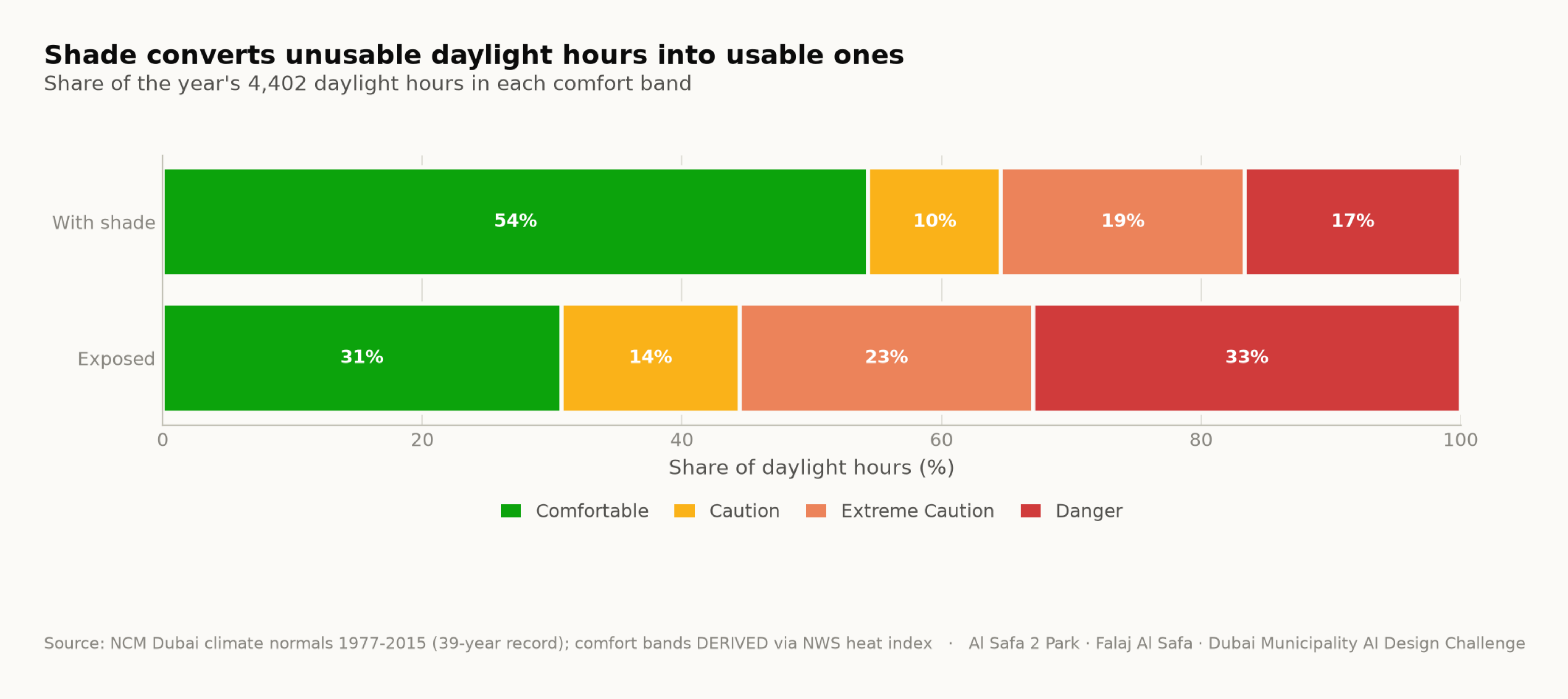
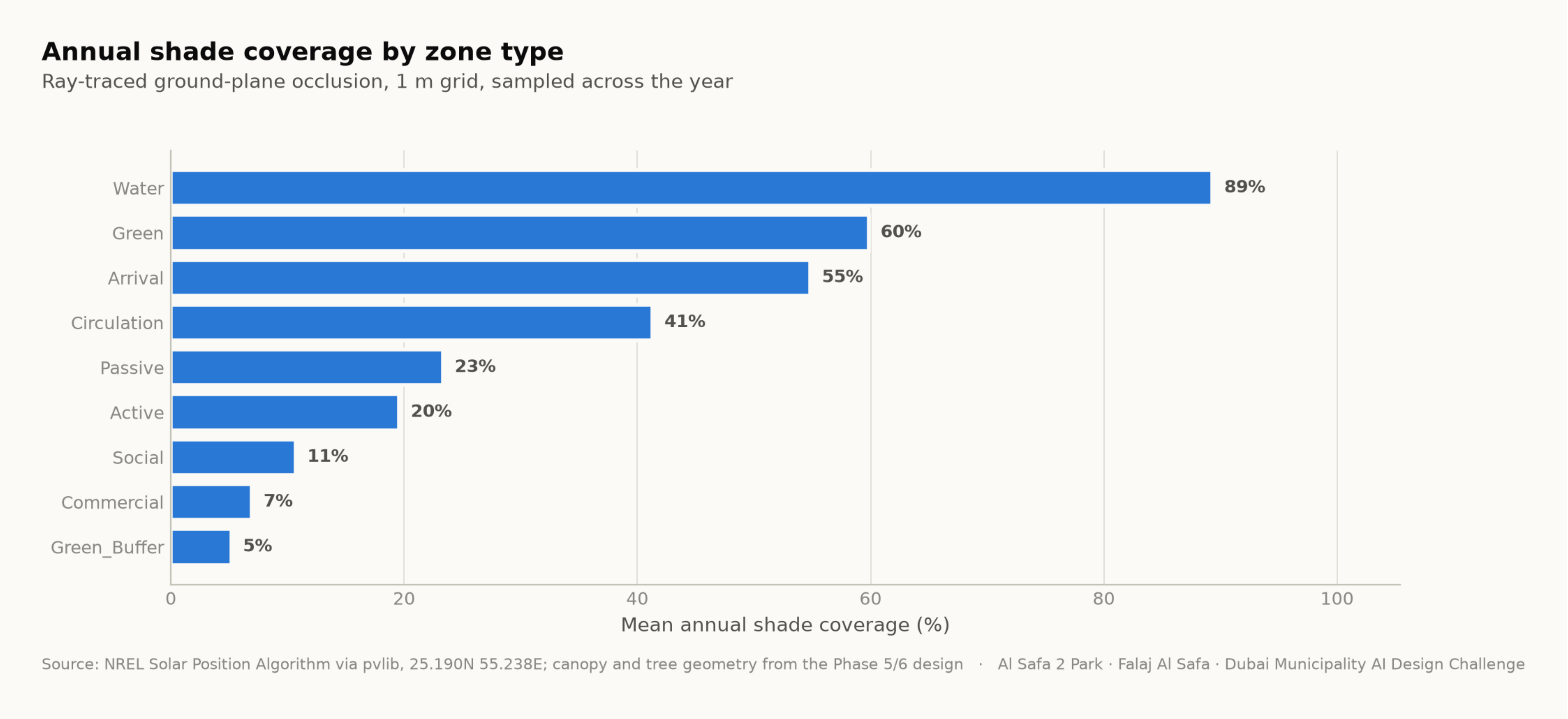


Fig03 shade by zone

Analysis output — computed from project data.



The complete project, and where to check it

Every one of the 129 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository	**github.com/wasim437/Al_SAFA**
Live portal	**wasim437.github.io/Al_SAFA/**

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (8)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
RENDER_PROMPTS.md	Project document
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/Al_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

tools/__init__.py	Build tool
tools/build_docs.py	Build tool
tools/build_links.py	Build tool
tools/build_notebook.py	Build tool
tools/build_reports.py	Build tool
tools/build_submission_pdfs.py	Build tool
tools/organize_repo.py	Build tool
tools/report_content.py	Build tool
tools/sync_film.py	Build tool
tools/sync_portal.py	Build tool
tools/sync_submission.py	Build tool

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset
data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Al Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD_PARKS_MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (9)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
archive/phases/02_PHASE2_PROBLEM_DEFINITION/README.md	Working record
archive/phases/03_PHASE3_OPPORTUNITY_AND_OBJECTIVES/README.md	Working record
archive/phases/04_PHASE4_CONCEPT_DEVELOPMENT/README.md	Working record
archive/phases/05_PHASE5_MASTERPLAN_DEVELOPMENT/README.md	Working record
archive/phases/06_PHASE6_DETAILED_DESIGN/README.md	Working record
archive/phases/07_PHASE7_PERFORMANCE_AND_SUSTAINABILITY/README.md	Working record
archive/phases/08_PHASE8_USER_EXPERIENCE_AND_ACTIVATION/README.md	Working record
archive/phases/09_PHASE9_AI_WORKFLOW_AND_VISUALIZATION/README.md	Working record

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